

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250275067

Kind Code

A1

Publication Date

August 28, 2025

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DISPLAY DEVICE

Abstract

A display apparatus can include a display panel having a non-bending area and a bending area adjacent to the non-bending area, and a support member located below the display panel and including a plurality of fiber bundles and a resin surrounding the plurality of fiber bundles. The plurality of fiber bundles are woven to have a first pattern in the non-bending area and are woven to have a second pattern in at least a partial area of the bending area. Further, at least one of a weave form or a fiber density of the first pattern is different from at least one of a weave form or a fiber density of the second pattern. As such, it is possible to prevent wrinkles even without a micro-patterning process when bending the display apparatus.

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Appl. No.: 19/028680

Filed: January 17, 2025

Foreign Application Priority Data

KR

10-2024-0029131

Feb. 28, 2024

Publication Classification

Int. Cl.: H05K5/02 (20060101); B32B5/02 (20060101); B32B15/04 (20060101); B32B17/06 (20060101); B32B27/06 (20060101); D03D1/00 (20060101); D03D13/00 (20060101)

U.S. Cl.:

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application claims priority to Korean Patent Application No. 10-2024-0029131, filed in the Republic of Korea on Feb. 28, 2024, the entire contents of which is hereby expressly incorporated by reference for all purposes as if fully set forth herein into the present application.

BACKGROUND

Field

[0002] Embodiments of the present disclosure relate to a display apparatus and a method for manufacturing the same.

Description of the Related Art

[0003] Recently, as the use of a display panel such as an LCD (liquid crystal display), an OLED (organic light emitting diode) and a PDP (plasma display panel) is generalized, attempts are being made to increase the ratio of an area where an image is implemented in an image display apparatus using the display panel.

[0004] In order to increase the ratio of an area where an image is implemented, the area of a bezel area where an image is not implemented needs to be reduced. However, since it is impossible to completely eliminate the bezel area, a limitation exists in increasing this ratio.

[0005] In this consideration, a display apparatus is being developed, in which, instead of reducing the size of a bezel area, a display panel is bent in the bezel area so that the bezel area is not visible.

[0006] However, when the display panel is bent, a problem can arise in that, due to a spring-back phenomenon, components included in the display apparatus can be separated from each other or bubbles can be generated.

[0007] Although a micro-patterning process is performed to prevent wrinkles that can occur in an area where the display panel is bent, a limitation is likely to be caused in that a process can be complicated and can take a long time.

SUMMARY OF THE DISCLOSURE

[0008] Embodiments of the present disclosure are directed to providing a display apparatus capable of preventing or minimizing wrinkles even without a micro-patterning process when bending the display apparatus.

[0009] Embodiments of the present disclosure are directed to providing a display apparatus capable of preventing or minimizing components included in the display apparatus from being separated from each other or bubbles from being generated due to a spring-back phenomenon when bending the display apparatus.

[0010] According to embodiments of the present disclosure, a display apparatus can include a display panel including a non-bending area and a bending area adjacent to the non-bending area; and a support member located below the display panel and including a plurality of fiber bundles and a resin surrounding the plurality of fiber bundles, wherein the plurality of fiber bundles are woven to have a first pattern in the non-bending area and are woven to have a second pattern in at least a partial area of the bending area and at least one of a weave form or a fiber density of the first pattern is different from at least one of a weave form or a fiber density of the second pattern.

[0011] According to embodiments of the present disclosure, a display apparatus can include a display panel including a first area and a second area adjacent to the first area; and a support

member disposed on a back surface of the display panel and including a plurality of fiber bundles, wherein the plurality of fiber bundles are woven in a first pattern in an area overlapping the first area and are woven in a second pattern in an area overlapping at least a part of the second area, wherein the first pattern and the second pattern are different weave type or the plurality of fiber bundles included in the first pattern and the plurality of fiber bundles included in the second pattern have different density.

[0012] According to the embodiments of the present disclosure, it is possible to provide a display apparatus capable of preventing or minimizing wrinkles even without a micro-patterning process when bending the display apparatus.

[0013] According to the embodiments of the present disclosure, it is possible to provide a display apparatus capable of preventing or minimizing components included in the display apparatus from being separated from each other or bubbles from being generated due to a spring-back phenomenon when bending the display apparatus.

[0014] According to the embodiments of the present disclosure, since a separate micro-patterning process for controlling wrinkles when bending a display apparatus is not required, it is possible to provide a display apparatus having process optimization.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0015] The present disclosure will become more fully understood from the detailed description given hereinbelow and the accompanying drawings which are given by way of illustration only, and thus are not limitative of the present disclosure.

[0016] FIG. 1 is a view illustrating an example of the cross-sectional structure of a display apparatus according to embodiments of the present disclosure.

[0017] FIG. 2 is a view illustrating a part of the cross-sectional structure illustrated in FIG. 1.

[0018] FIG. 3 is a view illustrating another example of the cross-sectional structure of the display apparatus according to the embodiments of the present disclosure.

[0019] FIG. 4 is a view three-dimensionally illustrating a part of the cross-sectional structure illustrated in FIG. 3.

[0020] FIG. 5 is a view illustrating an example of a cross-sectional structure taken along a line I-I' of FIG. 4.

[0021] FIG. 6 is a view illustrating an example of the planar structure of a configuration illustrated in (a) of FIG. 5.

[0022] FIG. 7 is a view illustrating another example of the cross-sectional structure taken along the line I-I' of FIG. 4.

[0023] FIG. 8 is a view illustrating an example of the planar structure of a configuration illustrated in (a) of FIG. 7.

[0024] FIG. 9 is a view illustrating an example of a cross-sectional structure taken along a line II-II' of FIG. 4.

[0025] FIG. 10 is a view illustrating an example of the planar structure of a configuration illustrated in (a) of FIG. 9.

[0026] FIG. 11 is a view illustrating another example of the cross-sectional structure taken along the line II-II' of FIG. 4.

[0027] FIG. 12 is a view illustrating an example of the planar structure of a configuration illustrated in (a) of FIG. 11.

[0028] FIGS. 13A and 13B are views illustrating examples of changes in the cross-sectional and planar structures of the display apparatus when the display apparatus is bent.

[0029] FIG. 14 is a view illustrating an example of a method for manufacturing a display apparatus

according to embodiments of the present disclosure.

DETAILED DESCRIPTION OF THE EMBODIMENTS

[0030] In the following description of examples or embodiments of the present disclosure, reference will be made to the accompanying drawings in which it is shown by way of illustration specific examples or embodiments that can be implemented, and in which the same reference numerals and signs can be used to designate the same or like components even when they are shown in different accompanying drawings from one another. Further, in the following description of examples or embodiments of the present disclosure, detailed descriptions of well-known functions and components incorporated herein will be omitted when it is determined that the description can make the subject matter in some embodiments of the present disclosure rather unclear. The terms such as “including,” “having,” “containing,” “constituting,” “make up of” and “formed of” used herein are generally intended to allow other components to be added unless the terms are used with the term “only.” As used herein, singular forms are intended to include plural forms unless the context clearly indicates otherwise.

[0031] Terms, such as “first,” “second,” “A,” “B,” “(A)” or “(B)” can be used herein to describe elements of the present disclosure. Each of these terms is not used to define essence, order, sequence, number of elements, etc., but is used merely to distinguish the corresponding element from other elements.

[0032] When it is mentioned that a first element “is connected or coupled to,” “contacts or overlaps,” etc. a second element, it should be interpreted that, not only can the first element “be directly connected or coupled to” or “directly contact or overlap” the second element, but a third element can also be “interposed” between the first and second elements, or the first and second elements can “be connected or coupled to,” “contact or overlap,” etc. each other via a fourth element. Here, the second element can be included in at least one of two or more elements that “are connected or coupled to,” “contact or overlap,” etc. each other.

[0033] When time relative terms, such as “after,” “subsequent to,” “next,” “before” and the like, are used to describe processes or operations of elements or configurations, or flows or steps in operating, processing, manufacturing methods, these terms can be used to describe non-consecutive or non-sequential processes or operations unless the term “directly” or “immediately” is used together.

[0034] In addition, when any dimensions, relative sizes, etc. are mentioned, it should be considered that numerical values for elements or features, or corresponding information (e.g., level, range, etc.) include a tolerance or error range that can be caused by various factors (e.g., process factors, internal or external impact, noise, etc.) even when a relevant description is not specified. Further, the term “can” fully encompasses all the meanings of the term “can.”

[0035] Hereinafter, various embodiments of the present disclosure will be described in detail with reference to accompanying drawings. All the components of each display apparatus according to all embodiments of the present disclosure are operatively coupled and configured.

[0036] FIG. 1 is a view illustrating an example of the cross-sectional structure of a display apparatus according to embodiments of the present disclosure.

[0037] Referring to FIG. 1, a display apparatus **100** can include a support layer **110**, a back plate **120**, a display panel **130**, a polarizer **140**, a cover glass **150** and an adhesive layer **160**.

[0038] The support layer **110** can play a role of supporting the display apparatus **100**. The support layer **110** can be formed of metal. However, the support layer **110** is not limited thereto. The support layer **110** can be formed of plastic. For example, the support layer **110** can be polyimide (PI).

[0039] The back plate **120** can be disposed on the support layer **110**. The back plate **120** can play a role of maintaining the rigidity of the display panel **130** and preventing foreign substances from being attached to the underside of the display panel **130**.

[0040] The back plate **120** can be formed of a material with high rigidity, such as plastic or metal,

to maintain the rigidity of the display panel **130**.

[0041] The display panel **130** can be disposed on the back plate **120**. The display panel **130** can be a layer in which a plurality of subpixels are formed and transistors for driving the plurality of subpixels and so on are formed. The display panel **130** can be an organic light emitting display panel (OLED), but is not limited thereto.

[0042] The polarizer **140** can be disposed on the display panel **130**. The polarizer **140** can prevent external light from being reflected inside the display apparatus **100**, thereby improving the reflective visibility of the display apparatus **100**. The polarizer **140** can be formed as a film with a polarizing function.

[0043] The cover glass **150** can be disposed on the polarizer **140**. The cover glass **150** can play a role of protecting elements inside the display panel **130**. The cover glass **150** can be formed of glass or plastic and can be transparent.

[0044] The adhesive layer **160** can be disposed between the support layer **110** and the back plate **120**, between the back plate **120** and the display panel **130**, between the display panel **130** and the polarizer **140**, and between the polarizer **140** and the cover glass **150**.

[0045] The adhesive layer **160** can play a role of adhering respective layers of the display apparatus **100** to each other. The adhesive layer **160** can be, for example, optically clear adhesive (OCA) or pressure sensitive adhesive (PSA), but is not limited to either one.

[0046] The adhesive layer **160** can be cured through a heat curing process. The heat curing process can proceed for 50 minutes at 110° C. or for 120 minutes at 80° C. However, the present disclosure is not limited thereto.

[0047] The display apparatus **100** can include a bending area BA and a non-bending area NBA.

[0048] The bending area BA means an area where at least a part of the display apparatus **100** is bent. The non-bending area NBA means an area where the display apparatus **100** is not bent.

[0049] The non-bending area NBA can include the center area of the display apparatus **100**. The bending area BA can be located on both sides of the non-bending area NBA and can include the outer area of the display apparatus **100**.

[0050] However, the present disclosure is not limited thereto. The bending area BA can include the center area of the display apparatus **100**, and the non-bending area NBA can be located on both sides of the bending area BA.

[0051] The display apparatus **100** can maintain a flat shape without being bent in the non-bending area NBA. The non-bending area NBA can also be referred to as a flat area.

[0052] The display apparatus **100** can be bent at least partially in the bending area BA. When the bending area BA includes the outer area of the display apparatus **100** as illustrated in FIG. 1, the display apparatus **100** can have a structure in which the outer area is bent in a direction parallel to a Z-direction as illustrated in FIG. 1.

[0053] The display panel **130** can include a display area in which a plurality of subpixels are disposed and a non-display area which surrounds the display area. The non-display area can also be referred to as a bezel area.

[0054] The display area can overlap the non-bending area NBA. The width in an X-direction of the display area can be equal to or larger than the width in the X-direction of the non-bending area NBA.

[0055] When the width in the X-direction of the display area is larger than the width in the X-direction of the non-bending area NBA, the display area can include the non-bending area NBA, and can partially overlap also the bending area BA.

[0056] The non-display area can overlap the bending area BA at the outer part of the display apparatus **100**. The width in the X-direction of the non-display area can be equal to or smaller than the width in the X-direction of the bending area BA.

[0057] In the case where the width in the X-direction of the non-display area is equal to or smaller than the width in the X-direction of the bending area BA, when the display apparatus **100** is bent in

the direction parallel to the Z-direction as illustrated in FIG. 1, a part of the display panel **130** where the non-display area is located can be entirely bent.

[0058] Therefore, when viewing the display apparatus **100** in the Z-direction, the non-display area may not be visible.

[0059] When implementing the display apparatus **100** in which both the outer areas are bent as illustrated in FIG. 1, because the non-display area is located within the bending area BA, by bending the display apparatus **100**, the non-display area can be caused to be invisible without reducing the width of the non-display area. Accordingly, the ratio of the display area, for example, an area where an image is implemented in the display apparatus **100**, can be maximized.

[0060] In order to implement the display apparatus **100** in which both outer areas are bent, it is necessary to maintain the bent state of the display apparatus **100** after the display apparatus **100** is bent.

[0061] FIG. 2 is a view illustrating a part of the cross-sectional structure illustrated in FIG. 1.

[0062] Referring to FIG. 2, the support layer **110** can include a plurality of grooves **200** in an area where the support layer **110** overlaps the bending area BA.

[0063] When the support layer **110** includes the plurality of grooves **200**, the support layer **110** can be bent more easily in an area where the plurality of grooves **200** are disposed than in an area where the plurality of grooves **200** are not disposed. In addition, as the plurality of grooves **200** are disposed, it is possible to prevent, to some extent, wrinkles from occurring in a process in which the display apparatus **100** is bent.

[0064] An adhesive material **210** can be further disposed under the support layer **110**. The adhesive material **210** can play a role of adhering materials located around the adhesive material **210** to each other. The adhesive material **210** can be the same material as the adhesive layer **160**, but is not limited thereto.

[0065] Both the adhesive material **210** and the adhesive layer **160** can be thermosetting materials.

[0066] When the display apparatus **100** is bent in the bending area BA, the adhesive material **210** can be absorbed into the plurality of grooves **200**.

[0067] When heat is applied to the adhesive material **210** absorbed into the plurality of grooves **200** for a specific time at a specific temperature, the adhesive material **210** can be cured. For example, in order to cure the adhesive material **210**, heat can be applied for 50 minutes at 110° C. or for 120 minutes at 80° C. However, the present disclosure is not limited thereto.

[0068] The adhesive force of the heat-cured adhesive material **210** can increase. When the adhesive force of the adhesive material **210** increases, the adhesive material **210** can prevent the support layer **110** from being deformed inside the plurality of grooves **200**. Accordingly, the support layer **110** can maintain a bent state in the bending area BA.

[0069] That is to say, as illustrated in FIG. 2, as the support layer **110** includes the plurality of grooves **200**, the display apparatus **100** can be easily bent in the bending area BA, and can maintain a bent structure by a process of heat-curing each of the adhesive layer **160** and the adhesive material **210**.

[0070] However, when the support layer **110** is formed of metal, a spring-back phenomenon can occur, in which the bent part of the support layer **110** is willing to return to its original flat shape due to restoring force.

[0071] When the spring-back phenomenon occurs, the support layer **110** does not maintain a bent structure and springs back to become straightened again.

[0072] However, since the adhesive layer **160** disposed on the support layer **110** still maintains a bent structure, the support layer **110** can be separated from the adhesive layer **160**. In addition, due to a separation space, bubbles can be generated inside the display apparatus **100**.

[0073] In other words, as components in the display apparatus **100** are separated from each other, a limitation can arise in relation with the reliability of the display apparatus.

[0074] In addition, in order to control wrinkles that are likely to occur when the display apparatus

100 is bent, the process of forming the plurality of grooves **200** in parts of the support layer **110** and attaching and heat-curing the separate adhesive material **210** as described above are necessary, but there is a disadvantage in that the process is complicated and takes a lot of time.

[0075] Hereinbelow, measures capable of solving these problems or limitations will be described with reference to the drawings.

[0076] FIG. **3** is a view illustrating another example of the cross-sectional structure of the display apparatus according to the embodiments of the present disclosure.

[0077] The cross-sectional structure of the display apparatus **100** illustrated in FIG. **3** is the same as the cross-sectional structure of the display apparatus **100** illustrated in FIG. **1** except that a support member **300** instead of the support layer **110** is disposed under the back plate **120** and there is no adhesive layer **160** under the back plate **120**, and thus, repeated description will be omitted or may be briefly provided.

[0078] Referring to FIG. **3**, the support member **300** can be disposed under the back plate **120**.

[0079] The support member **300** can be directly attached to the back plate **120**. The upper surface of the support member **300** can contact the lower surface of the back plate **120**.

[0080] The support member **300** can be a fiber reinforced plastic (FRP) prepreg which is made by impregnating a plurality of fiber bundles with a resin to surround the plurality of fiber bundles.

[0081] The plurality of fiber bundles can include glass fiber, carbon fiber or aramid fiber, but are not limited thereto.

[0082] The resin can include a urethane-based resin. Urethane resin can be heat-cured in a short time at a relatively low temperature compared to epoxy resin. For example, urethane resin can be heat-cured for a time equal to or shorter than **30** minutes at a temperature equal to or lower than **100° C**.

[0083] When the support member **300** includes urethane resin, since low-temperature curing and rapid curing are possible, there is an advantage in that heat forming can be performed after the support member **300** is attached to the back plate **120**.

[0084] The support member **300** can be a fabric type prepreg. In the fabric type prepreg, the plurality of fiber bundles can have a woven form. Namely, among the plurality of fiber bundles included in the support member **300**, fibers arranged in one direction and fibers arranged in another direction perpendicular to the one direction can be arranged to intersect each other. The plurality of fiber bundles included in the support member **300** can have different patterns in the non-bending area and the bending area. That means the plurality of fiber bundles are woven to have a first pattern in the non-bending area and are woven to have a second pattern in at least a partial area of the bending area. Thus, a different bending behavior can be achieved.

[0085] The different first and second pattern are created by different density and/or weaving of the fiber bundles.

[0086] So, the plurality of fiber bundles included in the support member **300** may have weave forms depending on a location. The fact that the plurality of fiber bundles included in the support member **300** have different weave forms depending on a location will be described later.

[0087] The support member **300** can further include a tackifier. The tackifier can increase the tack force of the support member **300**. The tackifier can be a compound, for example, such as cyclopentadiene, cyclopentene, styrene and indene, but is not limited thereto.

[0088] The tackifier can be mixed with the resin. In the process of manufacturing the support member **300** by impregnating the plurality of fiber bundles with the resin, the resin can be impregnated into the plurality of fiber bundles in the form of a mixture mixed with the tackifier.

[0089] The support member **300** including the tackifier can have a large tack force. For example, the tack force of the support member **300** in which the tackifier is included can be **25 gf/inch** or more.

[0090] Since the support member **300** has a tack force as the support member **300** includes the tackifier, the support member **300** can be attached to the back plate **120** without a separate adhesive

layer.

[0091] Since the support member **300** can be attached to the underside of the back plate **120** without a separate adhesive layer, the display apparatus **100** including the support member **300** can be manufactured with a thinner thickness.

[0092] FIG. **4** is a view three-dimensionally illustrating a part of the cross-sectional structure illustrated in FIG. **3**.

[0093] Referring to FIG. **4**, the support member **300** can include the bending area BA and the non-bending area NBA.

[0094] As described above, in the non-bending area NBA, the support member **300** is not bent and can maintain a flat shape. So, the central or middle portion is not bent.

[0095] The bending area BA can surround the non-bending area NBA on a plane that is defined by the X-direction and a Y-direction. The bending area BA in FIG. **4** fully surrounds the non-bending area NBA, however, it is not limited to that and it is also possible that the bending area BA only partly surrounds the non-bending area NBA, i.e. on one, two or only three side or on two opposing sides.

[0096] In the bending area BA, the support member **300** can be bent in a direction parallel to the Z-direction, i.e. up or down.

[0097] The bending area BA can include a plurality of first bending areas BA1 and a plurality of second bending areas BA2.

[0098] The first bending area BA1 can mean an area which is adjacent to an edge of the non-bending area NBA in the bending area BA.

[0099] The second bending area BA2 can mean an area which is adjacent to a point where edges of the non-bending area NBA intersect in the bending area BA. The second bending area BA2 can also be referred to as a corner area. So, the second bending area BA2 does not have a side adjacent to the non-bending area NBA. Only one corner of the second bending area BA2 is adjacent to a corner of the non-bending area NBA.

[0100] The plurality of fiber bundles included in the support member **300** can have different weave forms or different densities in the non-bending area NBA, the first bending area BA1 and the second bending area BA2, respectively.

[0101] Hereafter, various examples in which weave forms or densities of the plurality of fiber bundles included in the support member **300** are different will be described.

[0102] FIG. **5** is a view illustrating an example of a cross-sectional structure taken along a line I-I' of FIG. **4**. FIG. **6** is a view illustrating an example of the planar structure of a configuration illustrated in (a) of FIG. **5**.

[0103] Referring to FIGS. **5** and **6**, the plurality of fiber bundles included in the support member **300** can include fibers **501** arranged in the X-direction (hereinafter referred to as 'x1 fibers') and fibers **502** arranged in the Y-direction (hereinafter referred to as 'y1 fibers').

[0104] The support member **300** can be a single layer of prepreg which includes the plurality of fiber bundles and the resin.

[0105] Specifically, the x1 fibers **501** and the y1 fibers **502** can be arranged in one layer as illustrated in FIG. **5**. That is to say, all of the x1 fibers **501** can be arranged in the same layer, and all of the y1 fibers **502** can be arranged in the same layer.

[0106] The support member **300** can include areas where the x1 fibers **501** and the y1 fibers **502** are disposed to overlap each other and areas where the x1 fibers **501** and the y1 fibers **502** do not overlap each other.

[0107] The resin can be disposed to be filled between the plurality of fiber bundles.

[0108] Referring to (a) of FIG. **5**, the plurality of fiber bundles can have a plain weave form in the non-bending area NBA, and can have a satin weave form in the bending area BA.

[0109] The number of intersections of the fibers is less in the bending area BA compared to the non bending area.

[0110] In the non-bending area NBA, the x1 fibers **501** and the y1 fibers **502** can be disposed to intersect each other. A point where the x1 fiber **501** and the y1 fiber **502** intersect can appear every one fiber.

[0111] In the bending area BA, the x1 fibers **501** and the y1 fibers **502** can be disposed to intersect each other. A point where the x1 fiber **501** and the y1 fiber **502** intersect can appear every four fibers.

[0112] The x1 fibers **501** arranged in the bending area BA can be connected to the x1 fibers **501** arranged in the non-bending area NBA. Similarly, the y1 fibers **502** arranged in the bending area BA can be connected to the y1 fibers **502** arranged in the non-bending area NBA.

[0113] Referring to (b) of FIG. 5, the plurality of fiber bundles can have a twill weave form in the non-bending area NBA, and can have a satin weave form in the bending area BA.

[0114] As described above, in the non-bending area NBA, the x1 fibers **501** and the y1 fibers **502** can be disposed to intersect each other. A point where the x1 fiber **501** and the y1 fiber **502** intersect can appear every two fibers.

[0115] Thus, the number of intersections of the fibers is less in the bending area BA resulting in higher flexibility.

[0116] In the bending area BA, the x1 fibers **501** and the y1 fibers **502** can be disposed to intersect each other. A point where the x1 fiber **501** and the y1 fiber **502** intersect can appear every four fibers.

[0117] As described above, the x1 fibers **501** arranged in the bending area BA can be connected to the x1 fibers **501** arranged in the non-bending area NBA, and the y1 fibers **502** arranged in the bending area BA can be connected to the y1 fibers **502** arranged in the non-bending area NBA.

[0118] In FIG. 5, a case where the plurality of fiber bundles have a twill or plain weave form in the non-bending area NBA and have a satin weave form in the bending area BA is illustrated.

However, the present disclosure is not limited thereto, and the plurality of fiber bundles can have a plain or twill weave form in some areas of the bending area BA which are adjacent to the non-bending area NBA. Alternatively, the plurality of fiber bundles can have a satin weave form in some areas of the non-bending area NBA which are adjacent to the bending area BA.

[0119] FIG. 7 is a view illustrating another example of the cross-sectional structure taken along the line I-I' of FIG. 4. FIG. 8 is a view illustrating an example of the planar structure of a configuration illustrated in (a) of FIG. 7.

[0120] In the following description, content overlapping the content described above with reference to FIGS. 5 and 6 will be omitted or may be briefly provided.

[0121] Referring to FIGS. 7 and 8, the plurality of fiber bundles included in the support member **300** can include fibers **701** arranged in the X-direction (hereinafter referred to as 'x2 fibers') and fibers **702** arranged in the Y-direction (hereinafter referred to as 'y2 fibers').

[0122] Referring to (a) and (b) of FIG. 7, the plurality of fiber bundles can have the same weave form in the bending area BA and the non-bending area NBA. For example, the plurality of fiber bundles can have a plain weave form or a twill weave form in both the bending area BA and the non-bending area NBA.

[0123] Referring to (a) of FIG. 7 and FIG. 8, the x2 fibers **701** and the y2 fibers **702** can be arranged more densely in the non-bending area NBA than in the bending area BA. The distance between the x2 fibers **701** or the distance between the y2 fibers **702** can be larger in the bending area BA than in the non-bending area NBA.

[0124] The density of the plurality of fiber bundles included in the support member **300** can be smaller in the bending area BA than in the non-bending area NBA.

[0125] Specifically, in the non-bending area NBA, the support member **300** can have a porosity of 3% or less. A porosity can mean the ratio of an area in which the x2 fibers **701** and the y2 fibers **702** are not disposed in an area where the plurality of fiber bundles are disposed.

[0126] In the bending area BA, the support member **300** can have a porosity of 3% or more and

11% or less. In other words, an area where the x2 fibers **701** and the y2 fibers **702** are not disposed can be larger in the bending area BA than in the non-bending area NBA.

[0127] The x2 fibers **701** arranged in the bending area BA can be connected to the x2 fibers **701** arranged in the non-bending area NBA. Similarly, the y2 fibers **702** arranged in the bending area BA can be connected to the y2 fibers **702** arranged in the non-bending area NBA.

[0128] FIG. **9** is a view illustrating an example of a cross-sectional structure taken along a line II-II' of FIG. **4**. FIG. **10** is a view illustrating an example of the planar structure of a configuration illustrated in (a) of FIG. **9**.

[0129] In the following description, content overlapping the content described above with reference to FIGS. **5** to **8** will be omitted or may be briefly provided.

[0130] Referring to FIGS. **9** and **10**, the plurality of fiber bundles included in the support member **300** can include fibers **901** arranged in the X-direction (hereinafter referred to as 'x3 fibers') and fibers **902** arranged in the Y-direction (hereinafter referred to as 'y3 fibers').

[0131] The plurality of fiber bundles can have different weave forms in the non-bending area NBA, the first bending area BA1 and the second bending area (or the corner area) BA2.

[0132] Referring to (a) and (b) of FIG. **9**, the plurality of fiber bundles can have a plain or twill weave form in the non-bending area NBA, and can have a satin weave form in the first bending area BA1 and the second bending area BA2.

[0133] Among the plurality of fiber bundles, the fibers arranged in the first bending area BA1 can have a satin weave form in which the x3 fibers **901** and the y3 fibers **902** intersect every four fibers, and the fibers arranged in the second bending area BA2 can have a satin weave form in which the x3 fibers **901** and the y3 fibers **902** intersect every eight fibers.

[0134] The x3 fibers **901** arranged in the non-bending area NBA can be connected to the x3 fibers **901** arranged in the first bending area BA1. The x3 fibers **901** arranged in the first bending area BA1 can be connected to the x3 fibers **901** arranged in the second bending area BA2. Similarly, the y3 fibers **902** arranged in the non-bending area NBA can be connected to the y3 fibers **902** arranged in the first bending area BA1. The y3 fibers **902** arranged in the first bending area BA1 can be connected to the y3 fibers **902** arranged in the second bending area BA2.

[0135] In FIG. **9**, a case where the plurality of fiber bundles have a plain or twill weave form in the non-bending area NBA and a satin weave form in the first bending area BA1 and the second bending area BA2 is illustrated. However, the present disclosure is not limited thereto, and the plurality of fiber bundles can have a plain or twill weave form in some areas of the first bending area BA1 which are adjacent to the non-bending area NBA. Alternatively, the plurality of fiber bundles can have a satin weave form in some areas of the non-bending area NBA which are adjacent to the first bending area BA1.

[0136] Similarly, in some areas of the second bending area BA2 which are adjacent to the first bending area BA1, the plurality of fiber bundles can have a satin weave form in which the x3 fibers **901** and the y3 fibers **902** intersect every four fibers. Alternatively, in some areas of the first bending area BA1 which are adjacent to the second bending area BA2, the plurality of fiber bundles can have a satin weave form in which the x3 fibers **901** and the y3 fibers **902** intersect every eight fibers.

[0137] FIG. **11** is a view illustrating another example of the cross-sectional structure taken along the line II-II' of FIG. **4**. FIG. **12** is a view illustrating an example of the planar structure of a configuration illustrated in (a) of FIG. **11**.

[0138] In the following description, content overlapping the content described above with reference to FIGS. **9** and **10** will be omitted or may be briefly provided.

[0139] Referring to FIGS. **11** and **12**, the plurality of fiber bundles included in the support member **300** can include fibers **1101** arranged in the X-direction (hereinafter referred to as 'x4 fibers') and fibers **1102** arranged in the Y-direction (hereinafter referred to as 'y4 fibers').

[0140] Referring to (a) and (b) of FIG. **11**, the plurality of fiber bundles can have the same weave

form in the non-bending area NBA, the first bending area BA1 and the second bending area BA2. For example, all of the plurality of fiber bundles can have a plain weave form or a twill weave form in the first bending area BA1, the second bending area BA2 and the non-bending area NBA.

[0141] Referring to (a) of FIG. 11 and FIG. 12, the x4 fibers 1101 and the y4 fibers 1102 can be arranged more densely in the first bending area BA1 than in the second bending area BA2, and can be arranged more densely in the non-bending area NBA than in the first bending area BA1.

[0142] The distance between the x4 fibers 1101 or the distance between the y4 fibers 1102 can be larger in the first bending area BA1 than in the non-bending area NBA, and can be larger in the second bending area BA2 than in the first bending area BA1.

[0143] The density of the plurality of fiber bundles included in the support member 300 can be smaller in the first bending area BA1 than in the non-bending area NBA, and can be smaller in the second bending area BA2 than in the first bending area BA1.

[0144] Specifically, in the non-bending area NBA, the support member 300 can have a porosity of 3% or less. In the first bending area BA1, the support member 300 can have a porosity of 3% or more and 5% or less. In the second bending area BA2, the support member 300 can have a porosity of 5% or more and 11% or less. Namely, an area where the x4 fibers 1101 and the y4 fibers 1102 are not disposed can be larger in the first bending area BA1 than in the non-bending area NBA, and can be larger in the second bending area BA2 than in the first bending area BA1.

[0145] In the plurality of fiber bundles described with reference to FIGS. 5 to 12, the distance between intersections of intersecting fibers can be shortest in the non-bending area NBA.

Alternatively, x fibers and y fibers can be most densely arranged in the non-bending area NBA.

[0146] The distance between intersections of intersecting fibers can be longer in the bending area BA than in the non-bending area NBA. Alternatively, x fibers and y fibers can be arranged at a lower density in the bending area BA than in the non-bending area NBA.

[0147] In particular, in the second bending area BA2 of the bending areas BA which is adjacent to the area where the edges of the non-bending area NBA intersect, the distance between intersections of intersecting fibers can be longer than in the first bending area BA1. Alternatively, x fibers and y fibers can be arranged at a lower density in the second bending area BA2 than in the first bending area BA1.

[0148] As described with reference to FIGS. 5, 6, 9 and 10, when the distance between intersections of intersecting fibers in the bending area BA is larger than that in the non-bending area NBA, for example, when the plurality of fiber bundles have a satin weave form in the bending area BA, fibers can move more freely than when the plurality of fiber bundles have a plain or twill weave form. Accordingly, when the display apparatus 100 is bent, fibers can easily move to fit a bent shape, and thus, it is possible to prevent the shape of the support member 300 from being deformed due to bending.

[0149] Therefore, a limitation that wrinkles may occur when the display apparatus 100 is bent in the bending area BA can be solved, prevented or minimized.

[0150] Moreover, when the distance between intersections of intersecting fibers is longer in the corner area, for example, the second bending area BA2, than in the first bending area BA1, fibers can move more easily according to the structural deformation of the support member 300, and thus, a limitation that wrinkles may occur can be solved, prevented or minimized even in the corner area where wrinkles are most likely to occur when the display apparatus 100 is bent.

[0151] As described with reference to FIGS. 7, 8, 11 and 12, when the density of the plurality of fiber bundles is smaller in the bending area BA than in the non-bending area NBA, the disposition of fibers becomes loose and the distance between intersections of intersecting fibers becomes longer, and thus respective fibers can move more freely. Therefore, when the display apparatus 100 is bent, fibers can easily move to fit a bent shape, and thus, it is possible to prevent the shape of the support member 300 from being deformed when the display apparatus 100 is bent.

[0152] Therefore, a limitation that wrinkles may occur when the display apparatus 100 is bent in

the bending area BA can be solved, prevented or minimized.

[0153] Moreover, when the density of the plurality of fiber bundles is smaller in the corner area, for example, the second bending area BA2, than in the first bending area BA1, fibers can move more easily according to the structural deformation of the support member 300, and thus, a limitation that wrinkles may occur in the corner area can be solved, prevented or minimized. In addition, as described above, when using the support member 300 in which at least one of a weave form or the density of fibers in the bending area BA is different from a weave form or the density of fibers in the non-bending area NBA, the wrinkles of the display apparatus 100 can be controlled even without performing a micro-patterning process in which a plurality of grooves are formed in the support member 300 and an adhesive material is used and cured.

[0154] Furthermore, when the support member 300 including not metal but the plurality of fiber bundles and the resin is attached to the back plate 120, a spring-back phenomenon does not occur, and thus, it is possible to prevent or minimize a limitation that the support member 300 is separated from the back plate 120 or bubbles are generated in a separation space.

[0155] FIGS. 13A and 13B are views illustrating examples of changes in the cross-sectional and planar structures of the display apparatus when the display apparatus is bent.

[0156] Referring to FIG. 13A, the display apparatus 100 can include the support member 300 and an upper layer 1300 which is disposed on the support member 300.

[0157] The upper layer 1300 can include a back plate, a display panel, a polarizer, a cover glass and an adhesive layer.

[0158] The widths in the X-direction of the upper layer 1300 and the support member 300 in the bending area BA can both be d.

[0159] In the bending area BA, the upper layer 1300 and the support member 300 can be bent in the direction parallel to the Z-direction.

[0160] The upper layer 1300 can maintain a constant width in the bending area BA even when bending occurs. When the width of the upper layer 1300 in the bending area BA is constantly maintained, a change in optical performance due to bending can be minimized.

[0161] However, in order for the display apparatus 100 to have a bent structure even without the width of the upper layer 1300 changed as illustrated in FIG. 13A, the support member 300 should be easily compressible. That is to say, when bending occurs, a width d' of the support member 300 in the bending area BA can become smaller than the width d of the upper layer 1300 in the bending area BA.

[0162] In order for the support member 300 to be easily compressed and be changed in its width, fibers included in the support member 300 should be able to move freely.

[0163] Referring to FIG. 13B, the support member 300 can include the x2 fibers 701 and the y2 fibers 702.

[0164] Both the x2 fibers 701 and the y2 fibers 702 can have a plain weave form as described above as an example with reference to FIG. 8. However, the present disclosure is not limited thereto, both the x2 fibers 701 and the y2 fibers 702 can have a twill weave form. Alternatively, the x2 fibers 701 can have a plain or twill weave form, and the y2 fibers 702 can have a satin weave form.

[0165] As described above, the x2 fibers 701 and the y2 fibers 702 can be arranged more densely in the non-bending area NBA than in the bending area BA. The distance between the x2 fibers 701 or the distance between the y2 fibers 702 can be larger in the bending area BA than in the non-bending area NBA.

[0166] The support member 300 can be bent at least partially in the Z-direction in the bending area BA.

[0167] As the support member 300 is bent in the Z-direction, each of the distance between the x2 fibers 701 and the distance between the y2 fibers 702 can become smaller than before the support member 300 is bent. In other words, pores existing between the x2 fibers 701 or the y2 fibers 702

can be reduced.

[0168] As illustrated in FIG. 13B, since the distance between the x2 fibers 701 or the distance between the y2 fibers 702 is formed relatively longer in the bending area BA than in the non-bending area NBA, even when the support member 300 is bent, the x2 fibers 701 and the y2 fibers 702 in the bending area BA can move more freely than in the non-bending area NBA. For example, when bending occurs, the x2 fibers 701 and the y2 fibers 702 in the bending area BA can freely move to adjacent x2 fibers 701 and y2 fibers 702, or the lengths of the x2 fibers 701 and y2 fibers 702 in the bending area BA can easily decrease or increase compared to the x2 fibers 701 and y2 fibers 702 in the non-bending area NBA.

[0169] Since fibers located in the bending area BA can more easily move or be freely deformed such as by being stretched or shrinking compared to fibers located in the non-bending area NBA, even when bending occurs in the support member 300, wrinkles that can occur in the outer or corner area of the bending area BA can be effectively controlled.

[0170] Hereunder, a method for manufacturing a display apparatus according to embodiments of the present disclosure will be described.

[0171] FIG. 14 is a view illustrating an example of a method for manufacturing a display apparatus according to embodiments of the present disclosure.

[0172] Referring to (a) of FIG. 14, a plurality of fiber bundles can be woven to have a plain weave form in the non-bending area NBA and to have a satin weave form like (1) in the bending area BA or to have a plain weave form in the bending area BA but be arranged at a lower density in the bending area BA than fibers in the non-bending area NBA like (2).

[0173] In (a) of FIG. 14, a case where the plurality of fiber bundles have a plain weave form in the non-bending area NBA is illustrated. However, the present disclosure is not limited thereto, and the plurality of fiber bundles can have a twill weave form in the non-bending area NBA.

[0174] The plurality of fiber bundles which are woven can be impregnated with resin. The resin can include a urethane-based resin.

[0175] The resin can include a tackifier. As described above, the tackifier can be a compound such as cyclopentadiene, cyclopentene, styrene and indene, but is not limited thereto.

[0176] The resin including the tackifier can be impregnated into the plurality of fiber bundles which are woven.

[0177] Referring to (b) of FIG. 14, the plurality of fiber bundles impregnated with the resin or the support member 300 can be attached to the underside of the back plate 120 of the display apparatus 100.

[0178] Since the support member 300 has a tack force, the support member 300 can be directly attached to the lower surface of the back plate 120 without a separate adhesive layer.

[0179] After the support member 300 is directly attached to the lower surface of the back plate 120, the outer area of the display apparatus 100 can be removed by laser trimming.

[0180] Since a laser trimming process is performed after the support member 300 is attached to the back plate 120, the width in the X-direction of the support member 300 after the laser trimming process can be equal to the width in the X-direction of the back plate 120 or the width in the X-direction of the display panel 130.

[0181] Referring to (c) of FIG. 14, the laser-trimmed display apparatus 100 can be formed to be bent in the bending area BA.

[0182] The display apparatus 100 can be formed by a press process and a thermoforming process.

[0183] The press process can be a process in which the display apparatus 100 is inserted in molds with curved shapes and is then pressed from above and below the display apparatus 100 to have the same shape as the molds.

[0184] The thermoforming process can be performed simultaneously with the press process, and can be a process in which the resin included in the support member 300 is cured by applying heat.

[0185] The thermoforming process can proceed for a time equal to or shorter than 30 minutes at a

temperature equal to or lower than 100° C.

[0186] For example, the thermoforming process can proceed for 5 minutes at 100° C. or for 20 minutes at 90° C., but is not limited thereto.

[0187] The display apparatus **100** can have a bent shape in the bending area BA by the press process and the thermoforming process.

[0188] The resin included in the support member **300** is cured by the thermoforming process, and the tack force of the support member **300** increases due to curing of the resin. Thus, the support member **300** can be completely attached to the back plate **120**, allowing the display apparatus **100** to maintain a bent structure. For example, the fully cured resin can have a tack force equal to or greater than 1000 gf/inch.

[0189] The cover glass **150** can be formed on the display apparatus **100** with the bent shape.

[0190] The cover glass **150** can be attached to the display apparatus **100** in a shape bent in the same manner as the display apparatus **100**. The cover glass **150** can be attached to the polarizer **140** using an adhesive layer.

[0191] As the display apparatus **100** is attached to the cover glass **150** in a state in which the display apparatus **100** has the bent shape, it is possible to prevent wrinkles that can occur when attaching the flat display apparatus **100** to the cover glass **150** and then bending the flat display apparatus **100**.

[0192] As described above, the display apparatus **100** including the support member **300** can implement bending through a single thermoforming process.

[0193] On the other hand, in the display apparatus **100** described above with reference to FIG. 2, since both the adhesive layer **160** and the adhesive material **210** should be cured in order to implement bending, two thermoforming processes are required. Therefore, when the display apparatus **100** includes the support member **300**, a process can be simplified.

[0194] In addition, because the support member **300** does not require a separate adhesive layer, a separate attachment process is not required when connecting the back plate **120** and the support member **300**, and thus, the process can be further simplified.

[0195] Moreover, since the support member **300** includes a urethane-based resin, a curing time is shortened, and thus, process efficiency can be increased.

[0196] Brief description of the embodiments of the present disclosure described above is as follows.

[0197] According to embodiments of the present disclosure, a display apparatus can include a display panel including a non-bending area and a bending area which surrounds the non-bending area; and a support member located below the display panel and including a plurality of fiber bundles and a resin which surrounds the plurality of fiber bundles, wherein the plurality of fiber bundles are woven to have a first pattern in the non-bending area and are woven to have a second pattern in at least a partial area of the bending area and at least one of a weave form or a fiber density of the first pattern is different from at least one of a weave form or a fiber density of the second pattern.

[0198] In the display apparatus according to the embodiments of the present disclosure, the first pattern can be a plain weave form or a twill weave form, and the second pattern can be a satin weave form.

[0199] In the display apparatus according to the embodiments of the present disclosure, the first pattern and the second pattern can be a plain weave form or a twill weave form, and a density of the plurality of fiber bundles included in the second pattern can be lower than a density of the plurality of fiber bundles included in the first pattern.

[0200] In the display apparatus according to the embodiments of the present disclosure, the support member can be woven to have, in at least another partial area of the bending area, a third pattern different from at least one of the weave form or the fiber density of the second pattern.

[0201] In the display apparatus according to the embodiments of the present disclosure, the second

pattern can be located adjacent to an edge portion of the bending area, and the third pattern can be located adjacent to an intersection area of edge portions of the bending area.

[0202] In the display apparatus according to the embodiments of the present disclosure, at least one of a weave form or a fiber density of the third pattern can be different from at least one of the weave form or the fiber density of the first pattern.

[0203] In the display apparatus according to the embodiments of the present disclosure, the first pattern can be a plain weave form or a twill weave form and the second pattern and the third pattern can be a satin weave form, and the third pattern can have a longer length between intersections of the plurality of fiber bundles than the second pattern.

[0204] In the display apparatus according to the embodiments of the present disclosure, the first pattern, the second pattern and the third pattern can be a plain weave form or a twill weave form, and densities of the plurality of fiber bundles included in the first pattern, the second pattern and the third pattern can gradually decrease in the order of the first pattern, the second pattern and the third pattern.

[0205] In the display apparatus according to the embodiments of the present disclosure, the display apparatus can further include a back plate disposed between the display panel and the support member, wherein the support member can contact a lower surface of the back plate.

[0206] In the display apparatus according to the embodiments of the present disclosure, the support member can further include a tackifier.

[0207] In the display apparatus according to the embodiments of the present disclosure, the support member can be a single layer of prepreg.

[0208] In the display apparatus according to the embodiments of the present disclosure, a width in a horizontal direction of the support member can be equal to a width in the horizontal direction of the display panel.

[0209] In the display apparatus according to the embodiments of the present disclosure, the resin can include urethane resin.

[0210] According to embodiments of the present disclosure, a display apparatus can include a display panel including a first area and a second area which surrounds the first area; and a support member disposed on a back surface of the display panel and including a plurality of fiber bundles, wherein the plurality of fiber bundles are woven in a first pattern in an area overlapping the first area and are woven in a second pattern different from the first pattern in an area overlapping at least a part of the second area.

[0211] In the display apparatus according to the embodiments of the present disclosure, the first area can be flat, and at least a part of the second area can be bent.

[0212] In the display apparatus according to the embodiments of the present disclosure, the first pattern can be a plain weave form or a twill weave form, and the second pattern can be a satin weave form.

[0213] In the display apparatus according to the embodiments of the present disclosure, the first pattern and the second pattern can be a plain weave form or a twill weave form, and a density of the plurality of fiber bundles included in the second pattern can be lower than a density of the plurality of fiber bundles included in the first pattern.

[0214] In the display apparatus according to the embodiments of the present disclosure, the support member can be woven to have a third pattern different from the second pattern in an area overlapping at least another part of the second area.

[0215] In the display apparatus according to the embodiments of the present disclosure, the first pattern can be a plain weave form or a twill weave form and the second pattern and the third pattern can be a satin weave form, and the third pattern can have a longer length between intersections of the plurality of fiber bundles than the second pattern.

[0216] In the display apparatus according to the embodiments of the present disclosure, the first pattern, the second pattern and the third pattern can be a plain weave form or a twill weave form,

and densities of the plurality of fiber bundles included in the first pattern, the second pattern and the third pattern can gradually decrease in the order of the first pattern, the second pattern and the third pattern.

[0217] The above description has been presented to enable any person skilled in the art to make and use the technical idea of the present disclosure, and has been provided in the context of a particular application and its requirements. Various modifications, additions and substitutions to the described embodiments will be readily apparent to those skilled in the art, and the general principles defined herein can be applied to other embodiments and applications without departing from the spirit and scope of the present disclosure. The above description and the accompanying drawings provide an example of the technical idea of the present disclosure for illustrative purposes only. For example, the disclosed embodiments are intended to illustrate the scope of the technical idea of the present disclosure.

Claims

1. A display apparatus comprising: a display panel including a non-bending area and a bending area adjacent to the non-bending area; and a support member located below the display panel and including a plurality of fiber bundles and a resin surrounding the plurality of fiber bundles, wherein the plurality of fiber bundles are woven to have a first pattern in the non-bending area and are woven to have a second pattern in at least a partial area of the bending area, and wherein at least one of a weave form or a fiber density of the first pattern is different from at least one of a weave form or a fiber density of the second pattern.
2. The display apparatus of claim 1, wherein the first pattern is a plain weave form or a twill weave form, and the second pattern is a satin weave form.
3. The display apparatus of claim 1, wherein the first pattern and the second pattern are a plain weave form or a twill weave form, and wherein a density of the plurality of fiber bundles included in the second pattern is lower than a density of the plurality of fiber bundles included in the first pattern.
4. The display apparatus of claim 1, wherein the support member is woven to have a third pattern, in at least another partial area of the bending area, at least one of the weave form or the fiber density of the third pattern being different from at least one of the weave form or the fiber density of the second pattern.
5. The display apparatus of claim 4, wherein the second pattern is located adjacent to an edge portion of the bending area, and the third pattern is located adjacent to an intersection area of edge portions of the bending area.
6. The display apparatus of claim 4, wherein at least one of a weave form or a fiber density of the third pattern is different from at least one of the weave form or the fiber density of the first pattern.
7. The display apparatus of claim 4, wherein the first pattern is a plain weave form or a twill weave form, and the second pattern and the third pattern are a satin weave form, and wherein the third pattern has a longer length between intersections of the plurality of fiber bundles than the second pattern.
8. The display apparatus of claim 4, wherein the first pattern, the second pattern and the third pattern are a plain weave form or a twill weave form, and wherein densities of the plurality of fiber bundles included in the first pattern, the second pattern and the third pattern gradually decrease in the order of the first pattern, the second pattern and the third pattern.
9. The display apparatus of claim 1, further comprising: a back plate disposed between the display panel and the support member, wherein the support member contacts a lower surface of the back plate.
10. The display apparatus of claim 1, wherein the support member further includes a tackifier.
11. The display apparatus of claim 1, wherein a width in a horizontal direction of the support

member is equal to a width in the horizontal direction of the display panel.

12. The display apparatus of claim 4, wherein the at least another partial area of the bending area is a corner area.

13. The display apparatus of claim 1, wherein the plurality of fiber bundles include first fibers and second fibers, and wherein the first fibers arranged in the bending area are connected to the first fibers arranged in the non-bending area, and the second fibers arranged in the bending area are connected to the second fibers arranged in the non-bending area.

14. A display apparatus comprising: a display panel including a first area and a second area adjacent to the first area; and a support member disposed on a back surface of the display panel and including a plurality of fiber bundles, wherein the plurality of fiber bundles are woven in a first pattern in an area overlapping the first area and are woven in a second pattern in an area overlapping at least a part of the second area, wherein the first pattern and the second pattern are different weave type or the plurality of fiber bundles included in the first pattern and the plurality of fiber bundles included in the second pattern have different density.

15. The display apparatus of claim 14, wherein the first area is flat, and at least a part of the second area is bent.

16. The display apparatus of claim 14, wherein the first pattern is a plain weave form or a twill weave form, and the second pattern is a satin weave form.

17. The display apparatus of claim 14, wherein the first pattern and the second pattern are a plain weave form or a twill weave form, and wherein a density of the plurality of fiber bundles included in the second pattern is lower than a density of the plurality of fiber bundles included in the first pattern.

18. The display apparatus of claim 14, wherein the support member is woven to have a third pattern different from the second pattern in an area overlapping at least another part of the second area.

19. The display apparatus of claim 18, wherein the first pattern is a plain weave form or a twill weave form, and the second pattern and the third pattern are a satin weave form, and wherein the third pattern has a longer length between intersections of the plurality of fiber bundles than the second pattern.

20. The display apparatus of claim 18, wherein the first pattern, the second pattern and the third pattern are a plain weave form or a twill weave form, and wherein densities of the plurality of fiber bundles included in the first pattern, the second pattern and the third pattern gradually decrease in the order of the first pattern, the second pattern and the third pattern.
